Lesson 14: Accepting command line arguments

In C++ it is possible to accept command line arguments. To do so, you must first understand the full definition of int main(). It actually accepts two arguments, one is number of command line arguments, the other is a listing of the command line arguments.

It looks like this:

```
int main ( int argc, char *argv[] )
```

The integer, argc is the ARGument Count (hence argc). It is the number of arguments passed into the program from the command line, including the name of the program.

The array of character pointers is the listing of all the arguments. argv[0] is the name of the program, or an empty string if the name is not available. After that, every element number less than argc are command line arguments. You can use each argv element just like a string, or use argv as a two dimensional array. argv[argc] is a null pointer.

How could this be used? Almost any program that wants its parameters to be set when it is executed would use this. One common use is to write a function that takes the name of a file and outputs the entire text of it onto the screen.

```
#include <fstream>
#include <iostream>

using namespace std;

int main ( int argc, char *argv[] )
{
  if ( argc != 2 ) // argc should be 2 for correct execution
    // We print argv[0] assuming it is the program name
    cout<<"usage: "<< argv[0] <<" <filename>\n";
  else {
    // We assume argv[1] is a filename to open
    ifstream the_file ( argv[1] );
    // Always check to see if file opening succeeded
    if ( !the_file.is_open() )
      cout<<"Could not open file\n";
    else {
      char x;
      // the_file.get ( x ) returns false if the end of the file
      //  is reached or an error occurs
      while ( the_file.get ( x ) )
        cout<< x;
    }
    // the_file is closed implicitly here
  }
}
```

This program is fairly simple. It incorporates the full version of main. Then it first checks to ensure the user added the second argument, theoretically a file name. The program then checks to see if the file is valid by trying to open it. This is a standard operation that is effective and easy. If the file is valid, it gets opened in the process. The code is self-explanatory, but is littered with comments, you should have no

trouble understanding its operation this far into the tutorial. :-)